In [2]:
```
import pandas as pd
df = pd.read_csv("diabetes_prediction_dataset.csv")
df.head(5)
```

Out[2]:

| | gender | age | hypertension | heart_disease | smoking_history | bmi | HbA1c_level | blood_g |
|---|---|---|---|---|---|---|---|---|
| **0** | Female | 80.0 | 0 | 1 | never | 25.19 | 6.6 | |
| **1** | Female | 54.0 | 0 | 0 | No Info | 27.32 | 6.6 | |
| **2** | Male | 28.0 | 0 | 0 | never | 27.32 | 5.7 | |
| **3** | Female | 36.0 | 0 | 0 | current | 23.45 | 5.0 | |
| **4** | Male | 76.0 | 1 | 1 | current | 20.14 | 4.8 | |

In [3]:
```
df.tail(5)
```

Out[3]:

| | gender | age | hypertension | heart_disease | smoking_history | bmi | HbA1c_level | bl |
|---|---|---|---|---|---|---|---|---|
| **99995** | Female | 80.0 | 0 | 0 | No Info | 27.32 | 6.2 | |
| **99996** | Female | 2.0 | 0 | 0 | No Info | 17.37 | 6.5 | |
| **99997** | Male | 66.0 | 0 | 0 | former | 27.83 | 5.7 | |
| **99998** | Female | 24.0 | 0 | 0 | never | 35.42 | 4.0 | |
| **99999** | Female | 57.0 | 0 | 0 | current | 22.43 | 6.6 | |

In [4]:
```
print("Data type")
print(df.dtypes)
print("Data shape")
print(df.shape)
```

```
Data type
gender                  object
age                    float64
hypertension             int64
heart_disease            int64
smoking_history         object
bmi                    float64
HbA1c_level            float64
blood_glucose_level      int64
diabetes                 int64
dtype: object
Data shape
(100000, 9)
```

In [5]:
```
df['smoking_history'] = df['smoking_history'].replace({
    'former': 'never',
    'not current': 'never',
```

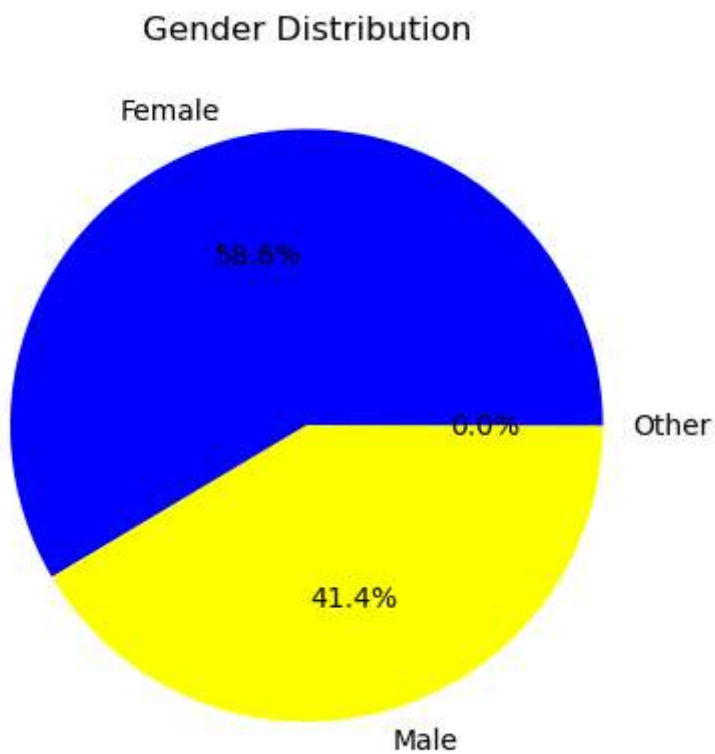
```
    'No Info': 'never'
})
df=df.dropna()
df.head(5)
```

Out[5]:

| | gender | age | hypertension | heart_disease | smoking_history | bmi | HbA1c_level | blood_g |
|---|---|---|---|---|---|---|---|---|
| **0** | Female | 80.0 | 0 | 1 | never | 25.19 | 6.6 | |
| **1** | Female | 54.0 | 0 | 0 | never | 27.32 | 6.6 | |
| **2** | Male | 28.0 | 0 | 0 | never | 27.32 | 5.7 | |
| **3** | Female | 36.0 | 0 | 0 | current | 23.45 | 5.0 | |
| **4** | Male | 76.0 | 1 | 1 | current | 20.14 | 4.8 | |

In [6]:

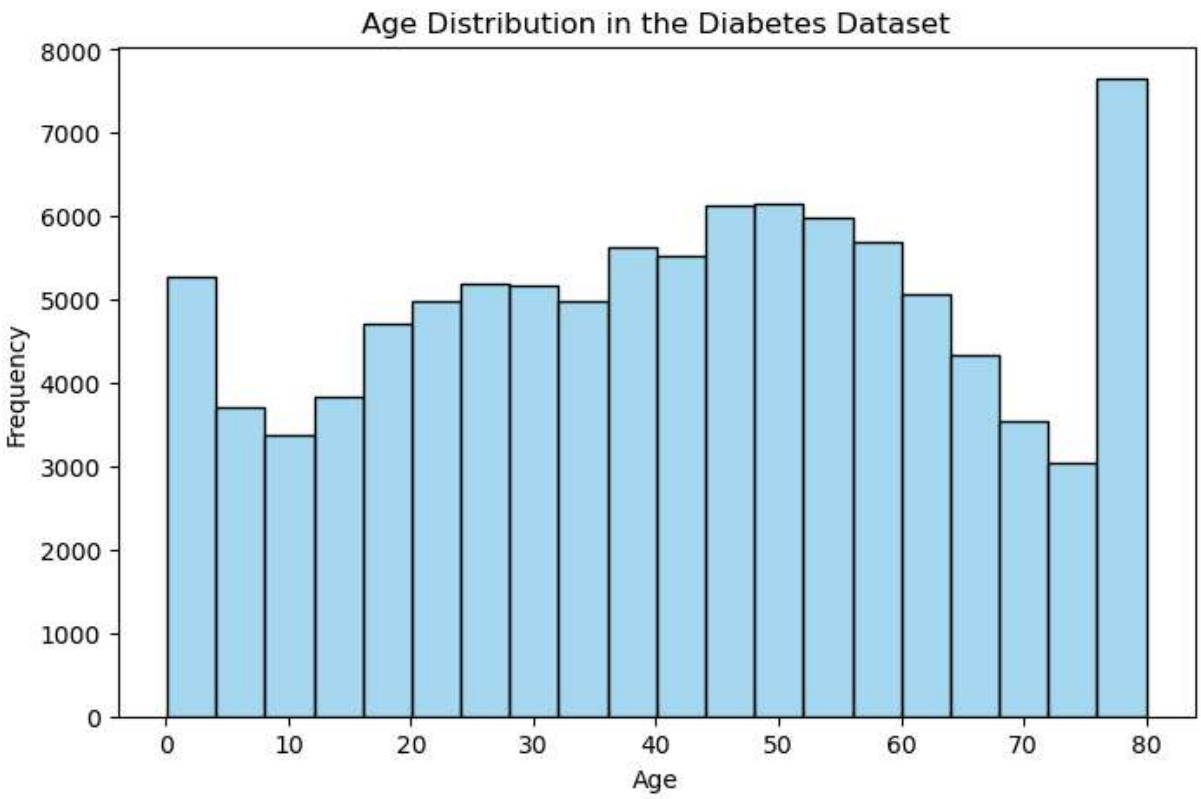
```
import matplotlib.pyplot as plt

gender_counts = df['gender'].value_counts()

plt.pie(gender_counts, labels=gender_counts.index,
        autopct='%1.1f%%', colors=['blue','yellow'])

plt.title("Gender Distribution")
plt.show()
```



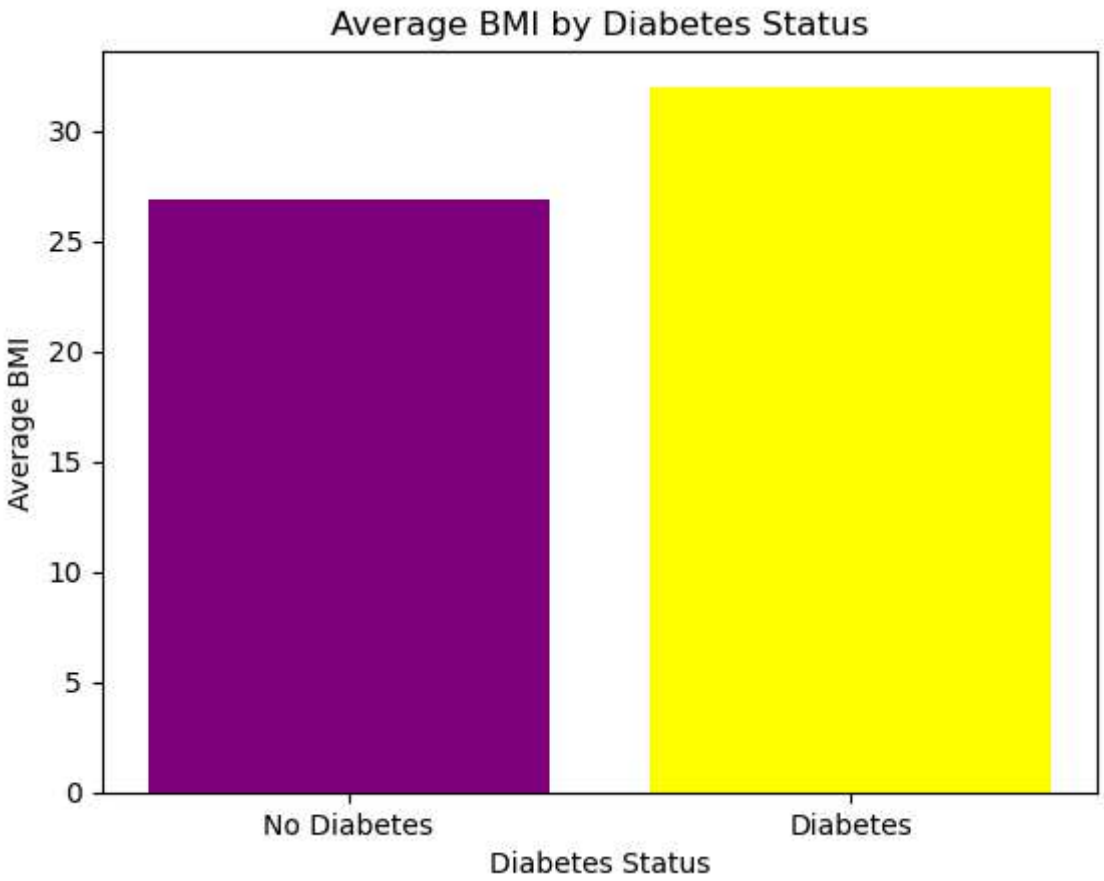

In [7]:

```
import seaborn as sns
import matplotlib.pyplot as plt
```

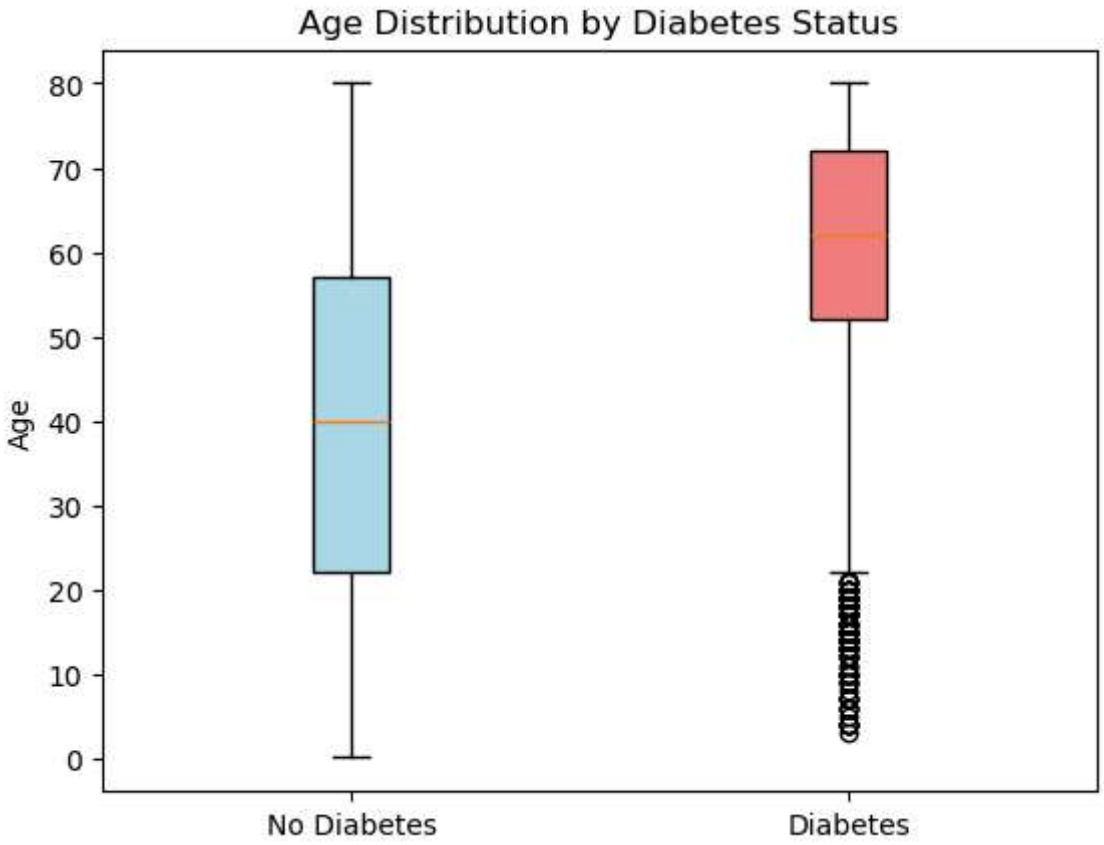
```
plt.figure(figsize=(8,5))
sns.histplot(
    data=df,
    x='age',
    bins=20,
    color='skyblue',
    edgecolor='black',
    kde=False
)

plt.title("Age Distribution in the Diabetes Dataset")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



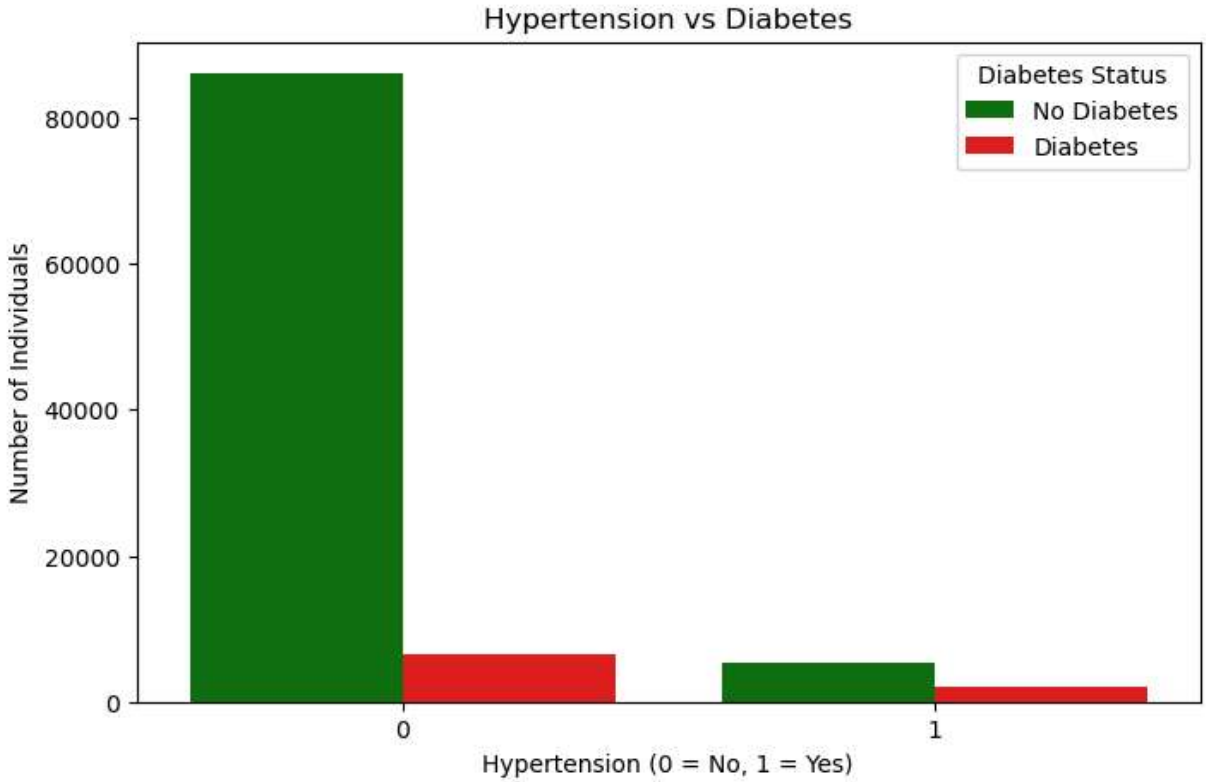

In [8]:
```
import matplotlib.pyplot as plt

bmi_diabetes = df.groupby('diabetes')['bmi'].mean()

plt.bar(['No Diabetes','Diabetes'], bmi_diabetes, color=['purple','yellow'])

plt.title("Average BMI by Diabetes Status")
plt.xlabel("Diabetes Status")
plt.ylabel("Average BMI")

plt.show()
```



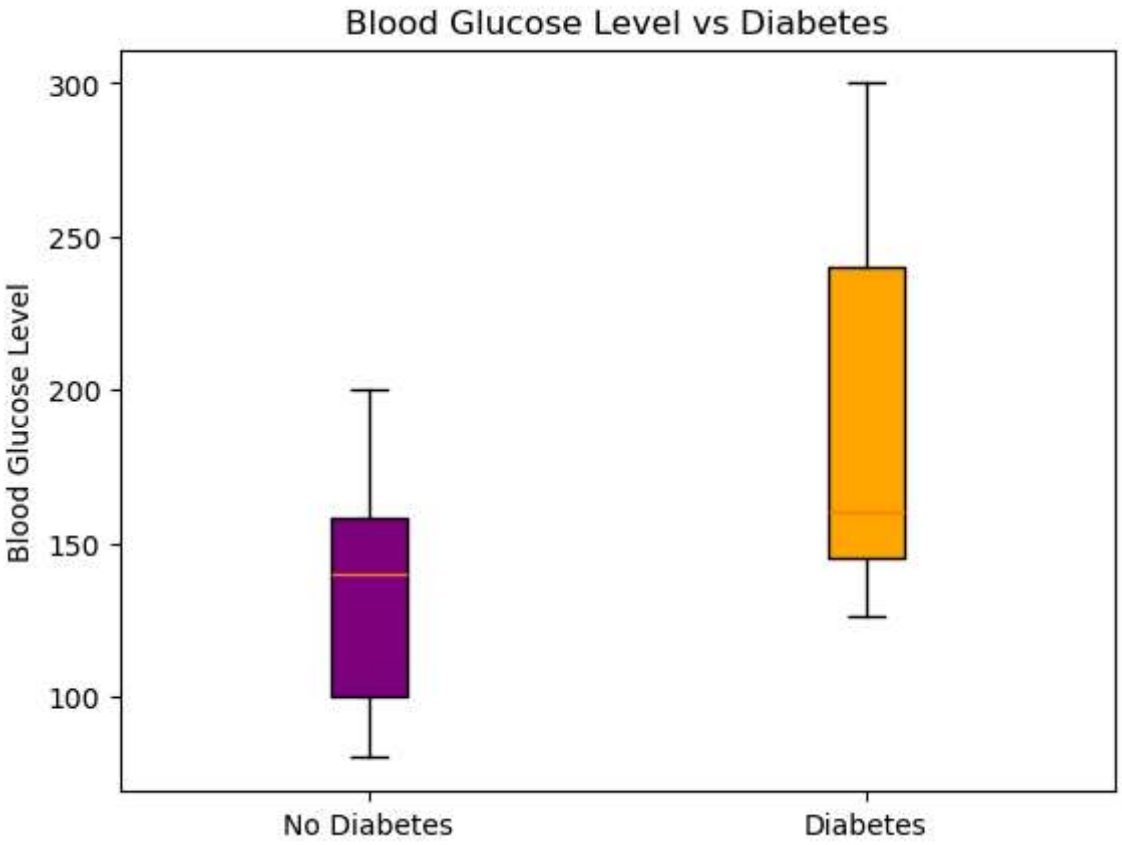

In [10]:
```
import matplotlib.pyplot as plt

box = plt.boxplot([df[df['diabetes']==0]['age'],
                   df[df['diabetes']==1]['age']],
                  tick_labels=['No Diabetes','Diabetes'],
                  patch_artist=True)

box['boxes'][0].set_facecolor('lightblue')
box['boxes'][1].set_facecolor('lightcoral')

plt.title("Age Distribution by Diabetes Status")
plt.ylabel("Age")

plt.show()
```



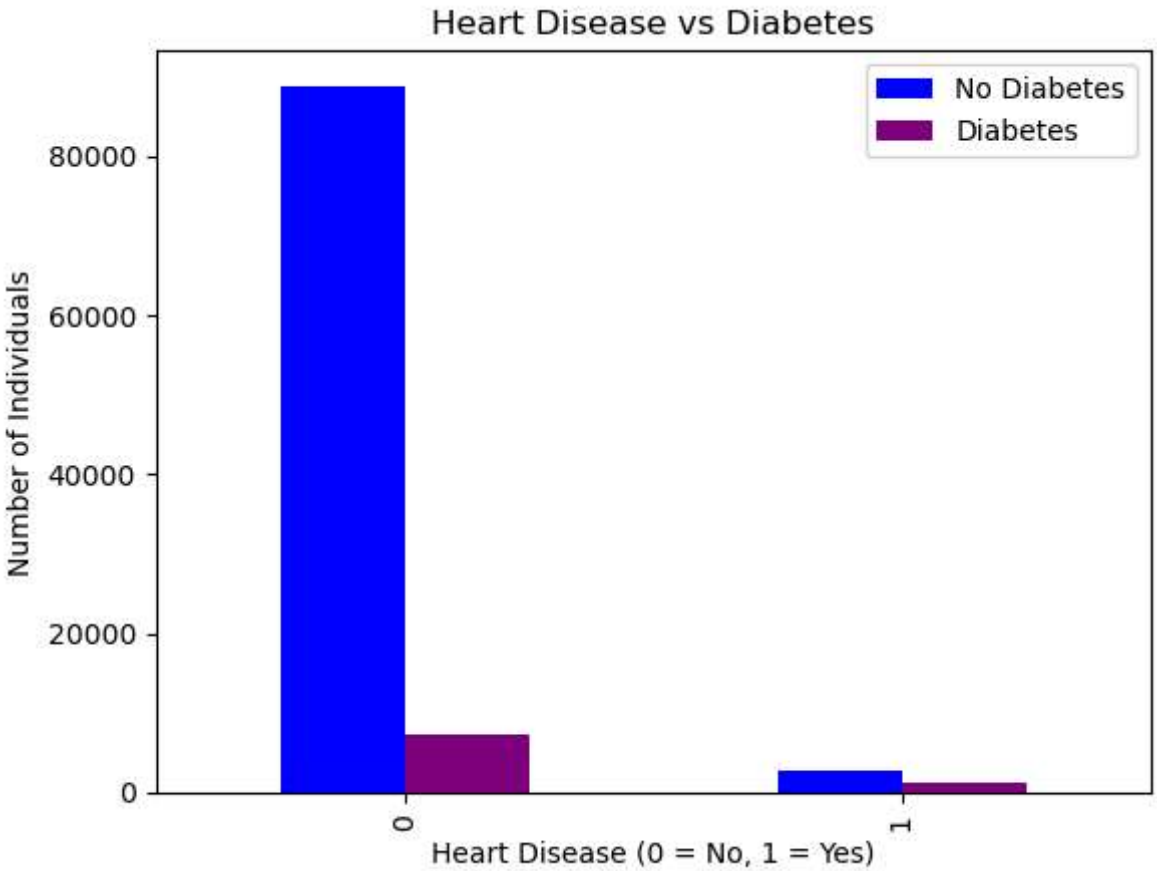

In [18]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
sns.countplot(
    data=df,
    x='hypertension',
    hue='diabetes',
    palette=['green','red']
)

plt.title("Hypertension vs Diabetes")
plt.xlabel("Hypertension (0 = No, 1 = Yes)")
plt.ylabel("Number of Individuals")
plt.legend(title="Diabetes Status", labels=["No Diabetes","Diabetes"])
plt.show()
```



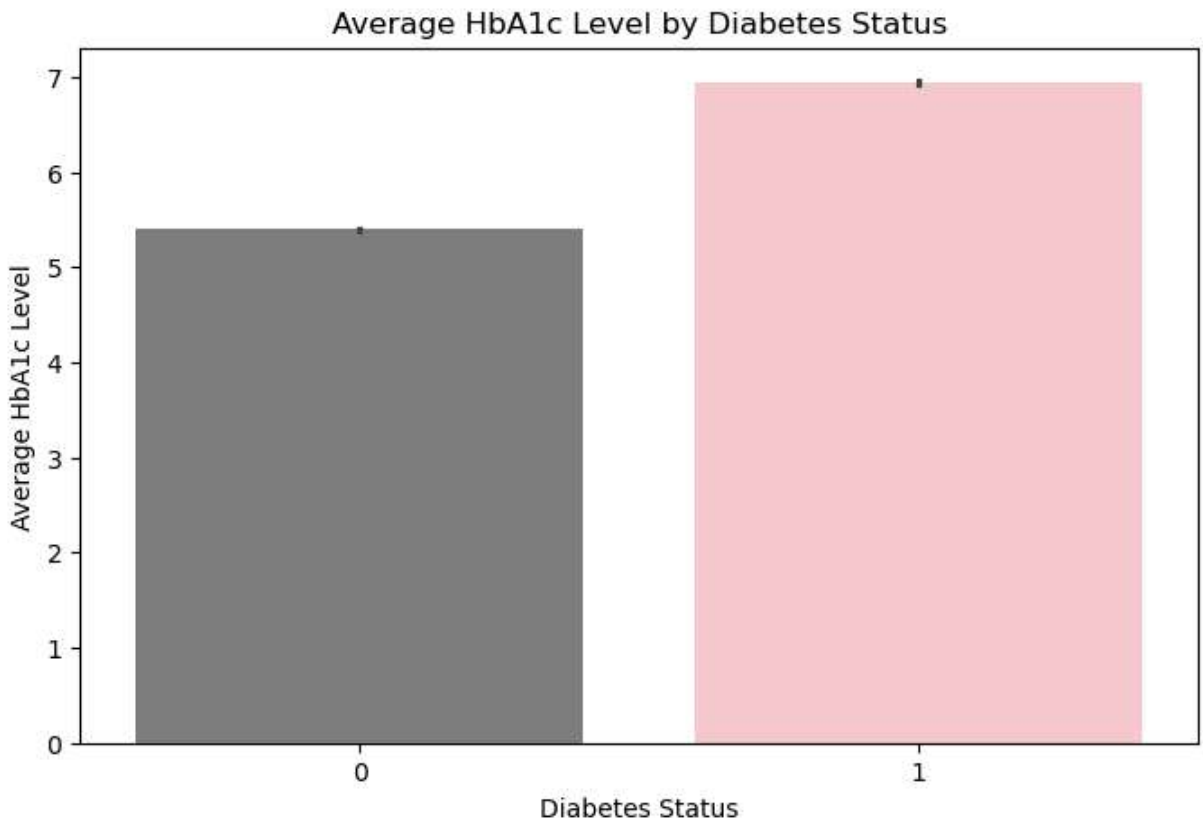

In [12]:
```
import matplotlib.pyplot as plt

box = plt.boxplot([df[df['diabetes']==0]['blood_glucose_level'],
                   df[df['diabetes']==1]['blood_glucose_level']],
                  tick_labels=['No Diabetes','Diabetes'],
                  patch_artist=True)

# add colors
box['boxes'][0].set_facecolor('purple')
box['boxes'][1].set_facecolor('orange')

plt.title("Blood Glucose Level vs Diabetes")
plt.ylabel("Blood Glucose Level")

plt.show()
```



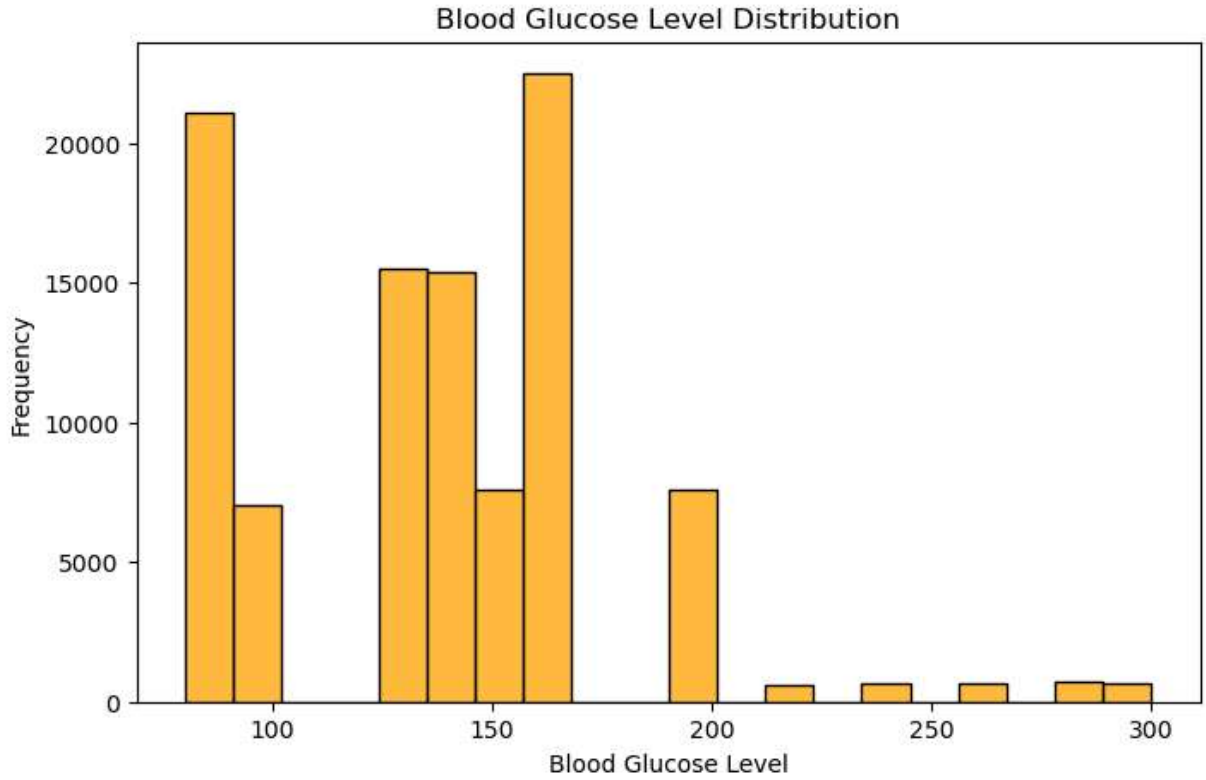

In [13]:
```
heart_diabetes = df.groupby(['heart_disease','diabetes']).size().unstack()

heart_diabetes.plot(kind='bar', color=['blue','purple'])

plt.title("Heart Disease vs Diabetes")
plt.xlabel("Heart Disease (0 = No, 1 = Yes)")
plt.ylabel("Number of Individuals")

plt.legend(['No Diabetes','Diabetes'])

plt.show()
```



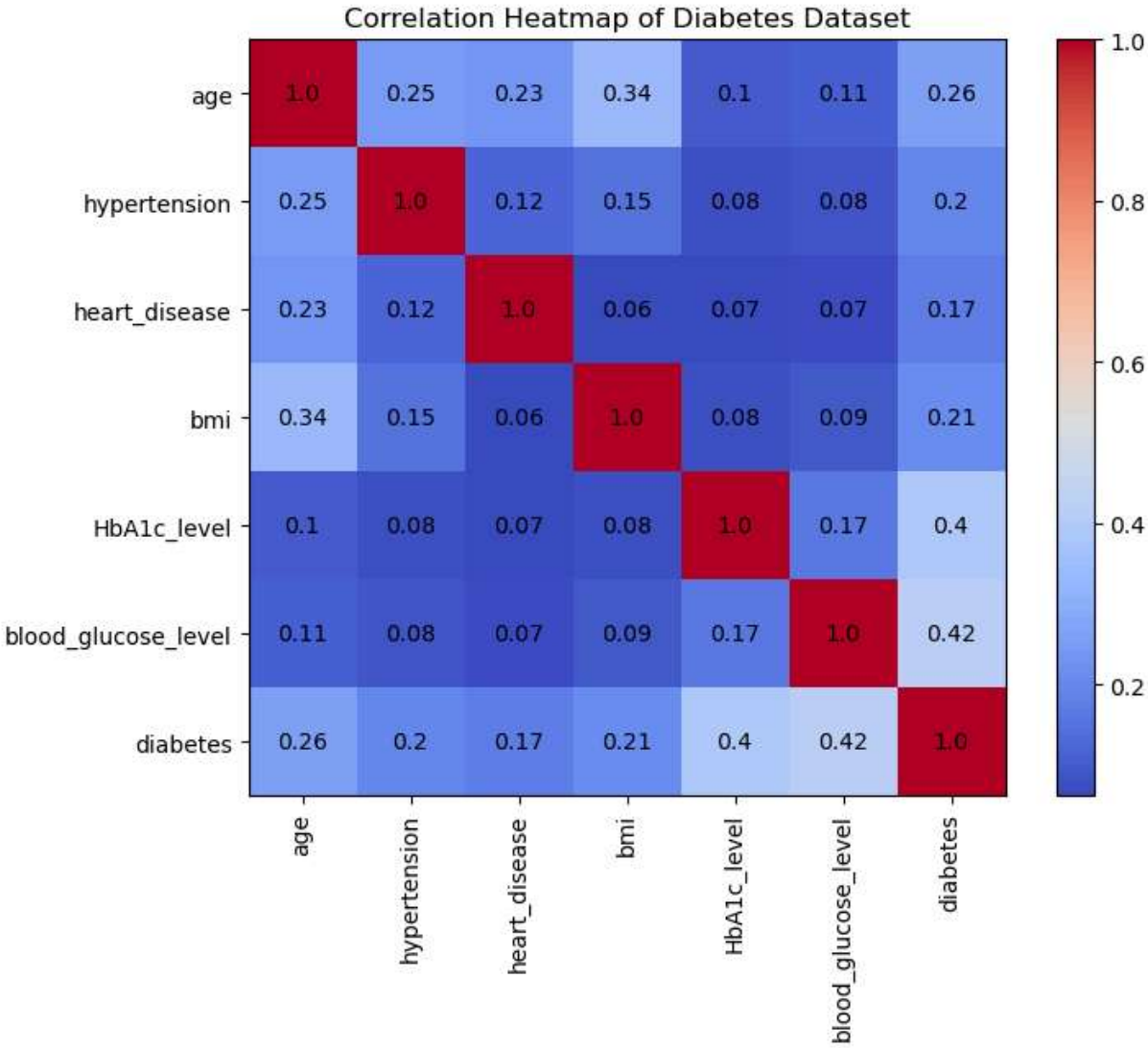

In [16]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
sns.barplot(
    data=df,
    x='diabetes',
    y='HbA1c_level',
    hue='diabetes',
    palette=['grey','pink'],
    dodge=False,
    legend=False
)

plt.title("Average HbA1c Level by Diabetes Status")
plt.xlabel("Diabetes Status")
plt.ylabel("Average HbA1c Level")
plt.show()
```



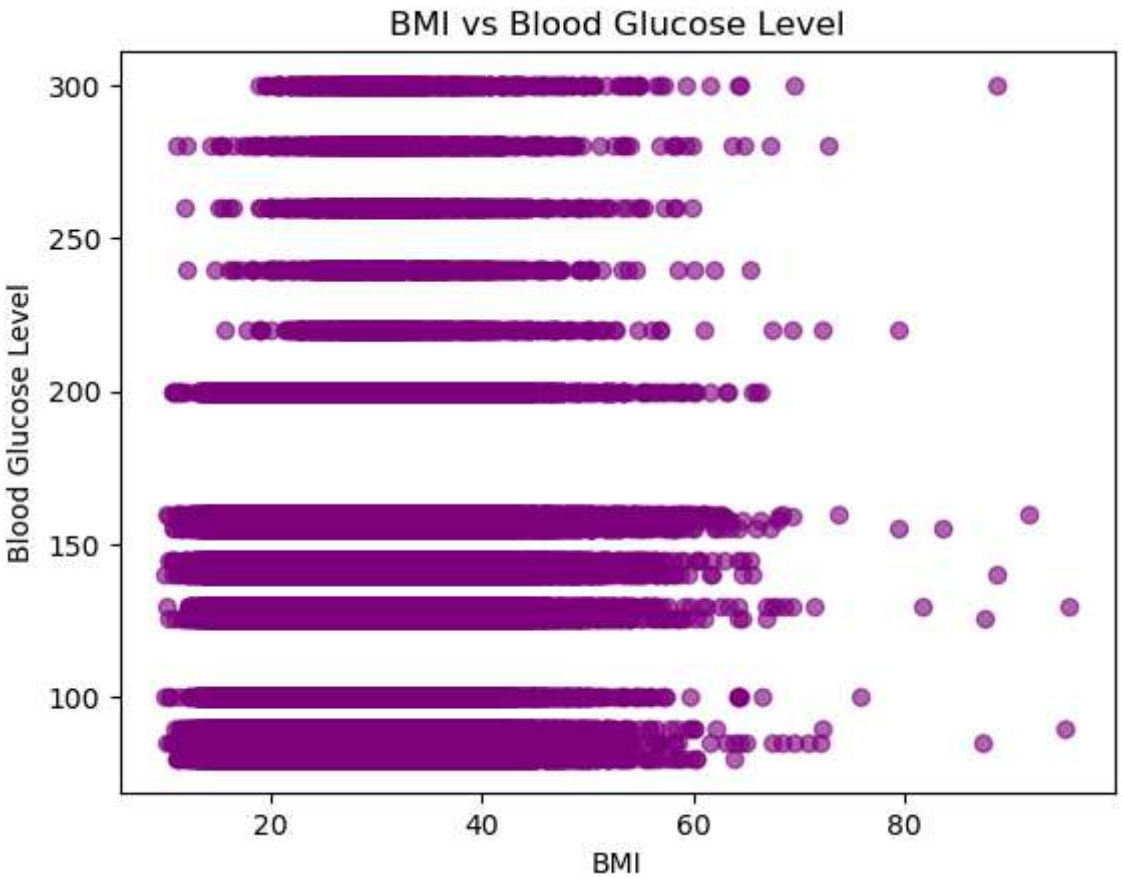

In [17]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
sns.histplot(data=df, x='blood_glucose_level', bins=20, color='orange', edgecolor='

plt.title("Blood Glucose Level Distribution")
plt.xlabel("Blood Glucose Level")
plt.ylabel("Frequency")
plt.show()
```



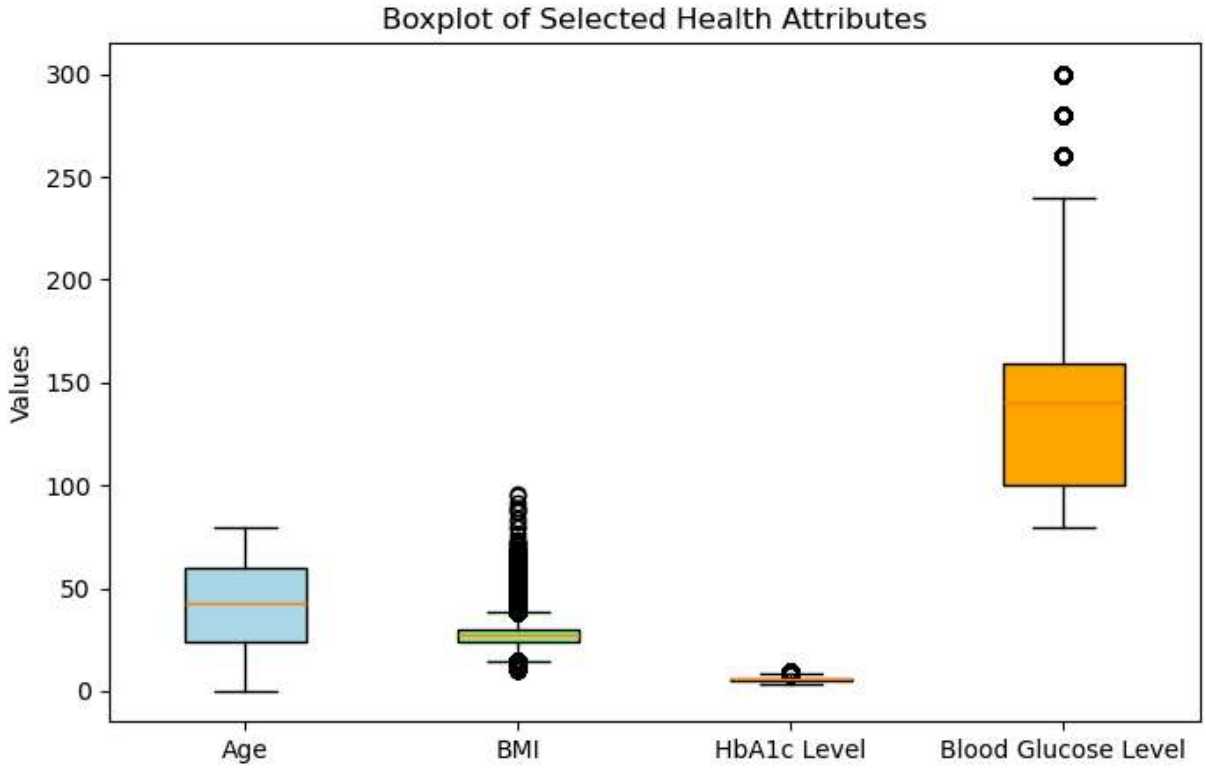

In [16]:
```
import matplotlib.pyplot as plt
import numpy as np

corr = df.corr(numeric_only=True)

plt.figure(figsize=(8,6))
plt.imshow(corr, cmap='coolwarm')

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)

# Add values inside boxes
for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(j, i, round(corr.iloc[i, j], 2),
                 ha='center', va='center', color='black')

plt.title("Correlation Heatmap of Diabetes Dataset")

plt.show()
```




In [19]:
```
plt.scatter(df['bmi'], df['blood_glucose_level'], color='purple', alpha=0.6)

plt.title("BMI vs Blood Glucose Level")
plt.xlabel("BMI")
plt.ylabel("Blood Glucose Level")

plt.show()
```




In [13]:
```
import matplotlib.pyplot as plt

data = [
    df['age'],
    df['bmi'],
    df['HbA1c_level'],
    df['blood_glucose_level']
]

labels = ['Age', 'BMI', 'HbA1c Level', 'Blood Glucose Level']

plt.figure(figsize=(8,5))

box = plt.boxplot(data, tick_labels=labels, patch_artist=True)

colors = ['lightblue', 'lightgreen', 'lightpink', 'orange']

for patch, color in zip(box['boxes'], colors):
    patch.set_facecolor(color)

plt.title("Boxplot of Selected Health Attributes")
plt.ylabel("Values")

plt.show()
```


Boxplot of Selected Health Attributes
Values
300
250
200
150
100
50
0
Age
BMI
HbA1c Level
Blood Glucose Level

In [ ]: